Explainable Analysis of Patient Profiles with Sleep Disorders: Clustering Approach and Identification of Key Factors

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Abstract

Sleep disorders have a major impact on patients, but their diagnosis remains complex due to the diversity of symptoms. Today, technological advances and the analysis of medical data offer new perspectives for better understanding them. In this work, we use real data from the *KANOPEE* application to analyze these disorders. Our objective is to better understand sleep disorders using artificial intelligence. In particular, *eXplainable Artificial Intelligence* (XAI), which aims to make AI decisions understandable and interpretable by users. We propose two methods: (1) a regression method to predict the average sleep duration, and (2) a method based on *clustering* to classify patients according to different disorder profiles. In both cases, an XAI approach is used to identify the key features influencing the target variable. A concrete application on anonymized real-world data demonstrates the relevance of our methods.

Keywords Sleep disorders · Explainable Artificial Intelligence (XAI) · Clustering · Statistical tests · Data analysis

1 Introduction

Sleep disorders become an important public health concern due to their harmful effects and consequences for patients, as highlighted in recent scientific studies [1]. Indeed, sleep disorders represent a major public health issue due to their multidimensional impact. These disorders affect not only people's quality of life, but also their mental and physical health, as well as their daily productivity.

Although early detection and appropriate treatment can significantly improve the health status of patients [2], their clinical management remains particularly complex due to the diversity of symptoms manifested and the heterogeneity of the associated causes [3]. A significant proportion of cases remain unrecognized and untreated. However, the emergence of digital health technologies [4] and the increased availability of medical data, be it from sleep tracking devices, medical records, or mobile apps, offer promising opportunities (perspectives) to improve the diagnosis of these disorders. As in many fields, AI is showing to be a promising tool for **detecting** and **analyzing** sleep disorders [3].

Alongside these advances, the complexity of AI-based decision support systems used in healthcare raises critical concerns about their transparency and reliability [5]. This led to a growing focus on explainable artificial intelligence (XAI), particularly in light of regulatory frameworks such as the European Union's General Data Protection Regulation (GDPR) [1]. Adopted in 2018, the GDPR emphasizes the rights of individuals to understand and challenge automated decisions made based on their data. As a result, there is a growing demand for interpretable models in critical areas such as digital health [5], driving research efforts to ensure that AI systems not only perform accurately but also provide understandable justifications for their outputs [5, 6].

In this work, we focus on the study of sleep disorders using real-world data from the *KANOPEE* application, dedicated to monitoring patients suffering from sleep-related disorders. Our objective is twofold: on the one hand, we seek to explain why an individual has a certain average sleep duration by developing a regression model capable of predicting this value. The results of this model are then interpreted using explainable artificial intelligence (XAI) techniques, in order to identify the variables that have the greatest influence on average sleep duration. By analyzing specific cases, we highlight the key factors that have the most significant impact on sleep time. On the other hand, we propose a method for classifying patients into homogeneous groups to identify different sleep disorder profiles. To achieve this, we apply the K-means clustering algorithm and characterize the obtained clusters using three complementary approaches: (i) statistical tests, (ii) distance-based measures, and (iii) an XAI analysis of the boundaries between clusters.

In what follows, we first describe the *KANOPEE* application and the collected data (Section 2). Next, we present our regression-based approach to explain individual sleep duration using predictive modeling and XAI techniques (Section 3), followed by a detailed presentation of our *clustering*-based method (Section 4). Finally, we discuss the results of the experimental protocol validating our approach (Section 5), before concluding the article and presenting future research directions (Section 6).

2 *KANOPEE* Data

We analyze data collected by Philip et al. [7] using *KANOPEE*, a smartphone application designed to reduce sleep complaints over a 17-day period through repeated interactions with a virtual assistant. KANOPEE [7] is Frances first virtual companion app dedicated to follow-up of patients with sleep disorders [2]. Developed and validated by Professor Pierre Philips team at *CHU de Bordeaux* and *CNRS SANPSY unit*.

KANOPEE provides digital support for patients suffering from sleep disorders via two virtual assistants (*Louise* and *Jeanne*). These agents carried out a personalized assessment during three visits (**D0-V0**, **D7-V1**, **D17-V2**) [3]. The questionnaires administered during these visits generate clinical indicators such as the **PHQ-9** (a depression scale) and the **ISI** (an insomnia severity scale).

The data includes information from three key interactions [7] : a **Screening Interview (V0)** on **Day 0 (D0)**, where users completed the **Insomnia Severity Index (ISI)**; a **Follow-Up Interview (V1)** on **Day 7 (D7)**, which included a summary of sleep diary entries, a second ISI assessment and personalized sleep recommendations; and a **Final Interview (V2)** on **Day 17 (D17)**, where users reported on their adherence to the recommendations, completed the ISI again, and were either advised to continue using KANOPEE autonomously (if ISI ≤ 21) or referred to a sleep specialist (if ISI > 21). To facilitate our analysis, the variables in the dataset were classified into three categories :

1. **Baseline data :** age, gender, BMI, location (department, region), socio-professional category, level of education, NOSAS score (risk of apnea), presence of OSA (sleep apnea syndrome) or RLS (restless legs syndrome).

2. **Measurements at visits (D-0, D-7, D-17) :** scores for anxiety (ANX), depression (DEP), insomnia (ISI), somnolence (ESS) and dates of visits.

3. **Derived variables :** means and standard deviations of sleep parameters (duration, sleep/wake hours) for D-0 $\rightarrow$ D-7 and D-7 $\rightarrow$ D-17.

One major problem was the presence of missing values, as most users did not complete all three visits. To guarantee the reliability of the analyses in a medical context, only complete records were retained, resulting in a dataset of **145 patients** and **37 features (i.e., data columns)**. In the following, we apply our two approaches.

[1] `https://eur-lex.europa.eu/eli/reg/2016/679/oj/eng`

[2] Information CHU de Bordeaux : Launch of Kanopée, the 1st virtual companion application to help with sleep issues, addiction, and stress related to lockdown, accessed: 2025-05-23.

[3] D: Day since registration, V: Visit

3 Regression-Based Approach to Explain Sleep Duration Using XAI

The first approach in our study aims to identify the variables that explain differences in sleep duration between individuals. The goal is to determine the factors that influence the number of hours of sleep by identifying those that may increase or decrease this duration, in order to better understand sleep disorders. To achieve this, we developed a regression model to predict the number of hours of sleep and used XAI techniques to explain the results using a set of representative instances. Based on these explanations, we draw conclusions about the variables that have the greatest influence on sleep duration. The method used is illustrated in Figure 1, and its steps are described below.

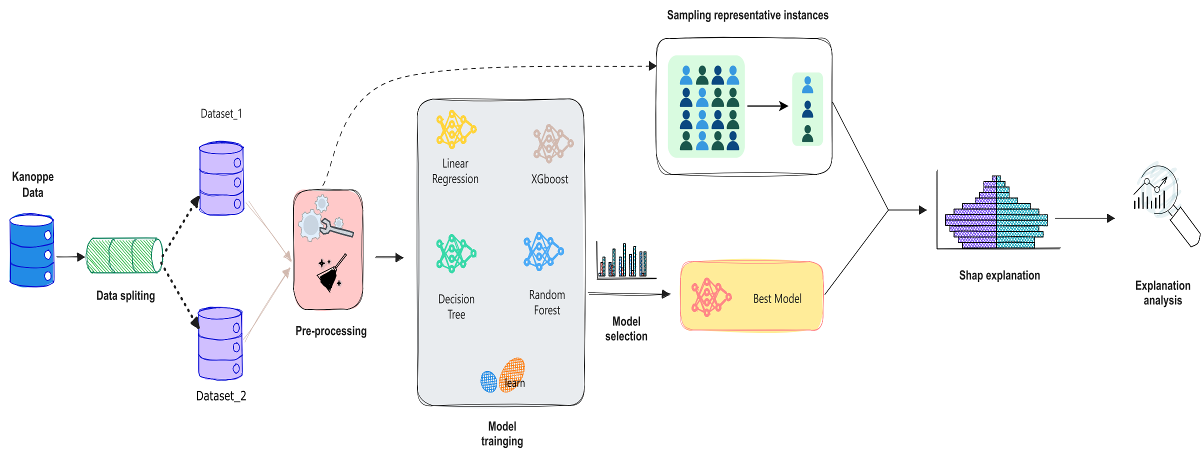

Figure 1: Methodology for identifying variables influencing sleep duration using predictive models

1. To train our regression models, we defined the average sleep duration as the target variable. Since the original dataset included two columns representing average sleep duration over two different time periods (D0–D7 and D7–D17)[4], we constructed two separate datasets, one for each period, by removing the column corresponding to the other period. This approach allowed us to train two independent models, each tailored to a specific timeframe.

2. After obtaining the two datasets, a preprocessing step was performed by encoding categorical variables using label encoding. Four regression models (Linear Regression, Decision Tree, Random Forest, and XGBoost) were then trained on each dataset, split into training (80%) and testing (20%) subsets. Their performance was evaluated using two metrics: Mean Squared Error (MSE) and Mean Absolute Error in minutes (MAE_{min}). The most accurate model was selected for the subsequent explainability step. The MAE in minutes is defined as $\text{MAE}_{\text{min}} = \frac{1}{N}\sum_{i=1}^{N} |(y_i - \hat{y}_i) \times 60|$, where y_i is the true value, $\hat{y}_i$ is the predicted value for the i-th instance, and N is the total number of instances. The factor 60 converts the error from hours to minutes.

3. To define the variables that affect sleep duration, we selected six representative instances from each dataset: two with average sleep duration, two with low values, and two with high values. These instances were excluded from the datasets prior to training the models. Using the model that achieved the best performance, we predicted the sleep duration for these instances and applied SHAP to explain the predictions, aiming to identify the variables that most influenced the predicted sleep duration for each case.

4. Based on the explanations of these instances and the SHAP summary plots, which highlight the features that most influence the prediction as well as the relationship between their values and their impact. Finally, we draw conclusions about the factors that affect sleep duration, identifying those that may increase or decrease this duration.

The first approach allows us to identify, in a general way, the key factors influencing sleep duration. While this provides valuable insights into overall trends, it does not fully account for the variability and individual differences that exist across the population. Indeed, symptoms and influencing factors can vary significantly from one group of individuals to another, due to differences in lifestyle, health status, or sociodemographic characteristics. This important observation motivates the development of our second approach, detailed in the following section, which focuses on segmenting

[4] `D0-D7` and `D7-D17` refer to Days 0–7 and 7–17, respectively.

patients into distinct groups using clustering techniques. By doing so, we are able to analyze sleep-related factors within each cluster, enabling a more personalized and nuanced interpretation of the data that better reflects the diversity of the studied population. This approach enables the analysis of sleep-related factors within each cluster, thereby allowing for a more personalized and nuanced interpretation of the data that more accurately captures the heterogeneity of the studied population.

4 Clustering and Characterization to Construct Patient Profiles

Our second approach consists of applying the *K-means* clustering algorithm to group patients according to their sleep profiles. The optimal number of clusters (K) was determined based on the highest silhouette score, which measures the quality of clustering. This choice was further supported by a Principal Component Analysis (PCA), allowing for visual confirmation of the cluster separability.

Three complementary methods were used to characterize the clusters and identify the influential variables:

- Comparative statistical tests between clusters [8]
- A distance-based analysis (*in-pattern/out-pattern*) [9]
- An explainable artificial intelligence (XAI) approach to interpret the boundaries between clusters, adapted from [10].

Figure 2 summarizes our methodological approach, with the applied methods described in the following subsections.

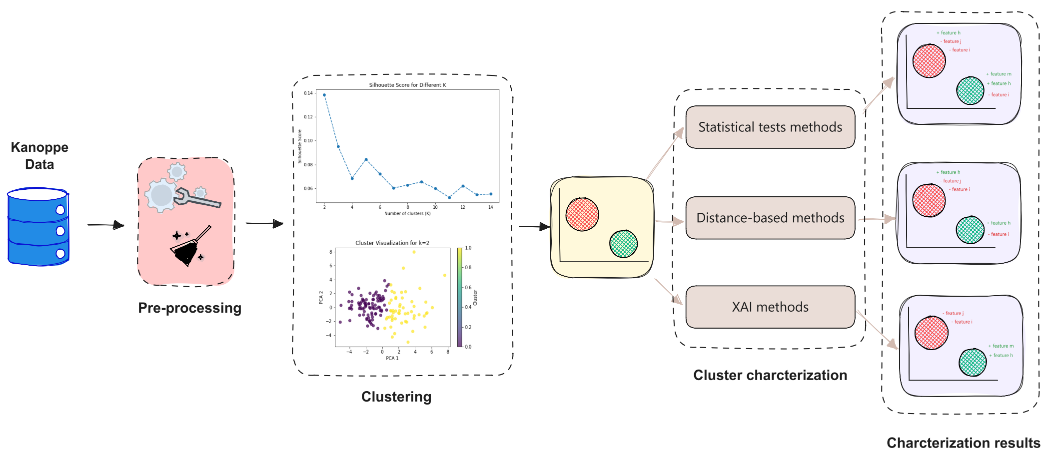

Figure 2: Illustrative diagram of the construction of patient profiles

Clustering K-means. Before applying the K-means algorithm[5], we performed a preprocessing step that included encoding categorical variables using label encoding and standardizing the numerical features. Standardization was carried out by transforming each feature x using the formula: $x' = \frac{x-\mu}{\sigma}$, where μ is the mean and σ is the standard deviation of the feature.

The clustering process resulted in the formation of distinct patient groups. Once the clusters were established, we applied three complementary methods to characterize them, with the goal of identifying the most salient variables specific to each group.

4.1 Comparative Statistical Tests

This method is inspired by the work of [8]. For numerical variables, the mean of each variable within each cluster is calculated, and statistical tests are applied to assess whether there is a significant difference between these means. First, an *F-test* is performed to determine whether the two populations have equal variances. If the variances are equal, the **Student's** t-test is applied; else, the **Welch** t-test is used. Only variables with a p-value ≤ 0.05 are retained as

[5]The optimal number of clusters in our case study is $K = 2$

significant characteristics, indicating a significant difference between clusters. For categorical variables, percentages by category within each cluster are compared. When there are only two categories (resulting in a 2×2 contingency table), the **Fisher's exact** test is used; otherwise, the **Chi-square** test is applied. Only variables with a p-value ≤ 0.05 are retained.

Once the characteristic variables have been identified[6], we interpret their average values cluster by cluster. For clinical indicators(`ANX, DEP, ISI, ESS, BMI`), whose standardized scales allow absolute interpretation, we analyze the means with reference to these norms. For other numerical variables, we perform a relative comparison between clusters. For categorical variables, we compare the percentages of each category within and between the two clusters. This detailed analysis ultimately leads to a comprehensive characterization of each group.

4.2 In-pattern/Out-pattern Method

The In-pattern/Out-pattern method is directly inspired by the work of [9], which proposes an approach that divides the observations into two groups: in-pattern and out-pattern. This analysis helps identify the distinctive variables for each cluster, as well as their trends (i.e., whether their values tend to be high or low). The main steps of the method are as follows:

1. **Centroid calculation**: Determine X_k as the mean of the points in cluster k.
2. **In/out-pattern classification**: For each point i, compute its Euclidean distance d_i to the centroid X_k. If $d_i \in [\mu_k \pm z \cdot \sigma_k]$, it is considered *in-pattern*; otherwise, it is *out-pattern*, where μ_k and σ_k are the mean and standard deviation of cluster k, and z is a threshold constant.
3. **In/out-pattern means**: Compute $\mu_{\text{in}}(k,v)$ and $\mu_{\text{out}}(k,v)$ for each cluster k and variable v.
4. **Differentiation factor calculation**: This step involves computing, for each variable v and each cluster k, a differentiation factor:
$$df(k,v) = \frac{\mu_{\text{in}}(k,v) - \mu_{\text{out}}(k,v)}{\mu_{\text{out}}(k,v)}$$
This factor measures the relative difference between the in-pattern and out-pattern means, allowing the identification of the most distinctive variables.
5. **Descriptive statistics**: Compute the means (μ) and standard deviations (σ) per cluster to characterize the distribution of each variable v within the cluster.
6. **Identification of salient variables**: A variable v is considered salient in cluster k if its differentiation factor satisfies:
$$\begin{aligned} & df(k,v) \leq \mu_{df}(k) - z\sigma_{df}(k) \\ \text{or} \quad & df(k,v) \geq \mu_{df}(k) + z\sigma_{df}(k) \end{aligned}$$
A positive factor indicates that v tends to take high values in cluster k, while a negative factor means its values are predominantly low.

This method identifies the characteristic variables of each cluster and indicates whether their values are predominantly high or low, thereby enabling a clear characterization of each group.

4.3 XAI-based Method

Based on the work of [10], this method applies XAI techniques to analyze and interpret clusters. The overall framework, depicted in Figure 3, is structured around the following four steps:

(a) We apply a clustering algorithm and assign to each instance a label corresponding to the cluster it belongs to (the target variable is denoted `Cluster`).

(b) We then train classifiers using 75% of the labeled data for training and 25% for testing. To ensure reliable and robust results, we used four models: **Logistic Regression**, **Decision Tree**, **Random Forest**, and **XGBoost**. For each cluster, we selected the most effective model based on accuracy and F_1 score.

[6] Variables showing significant differences between clusters

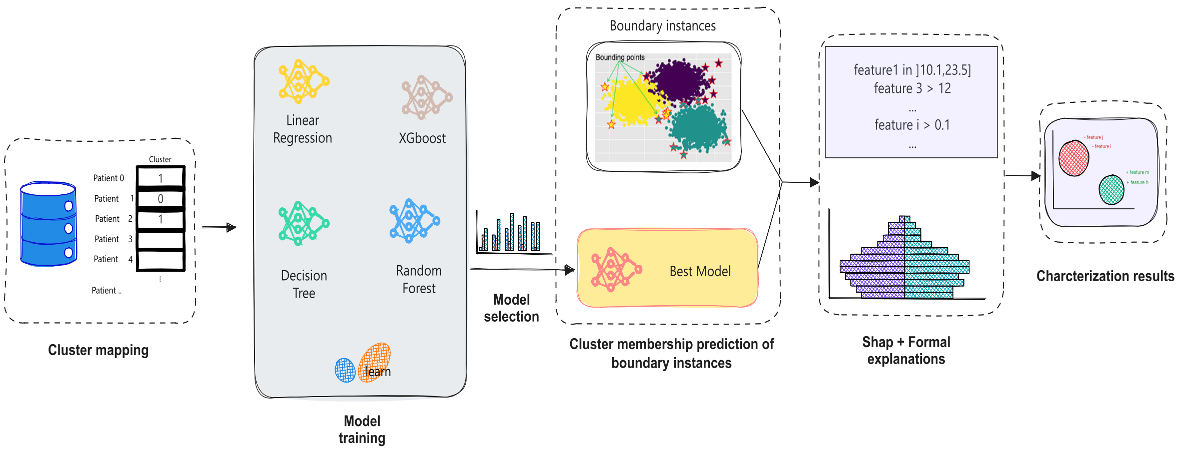

Figure 3: Methodological diagram for extracting characteristic variables (XAI-approach)

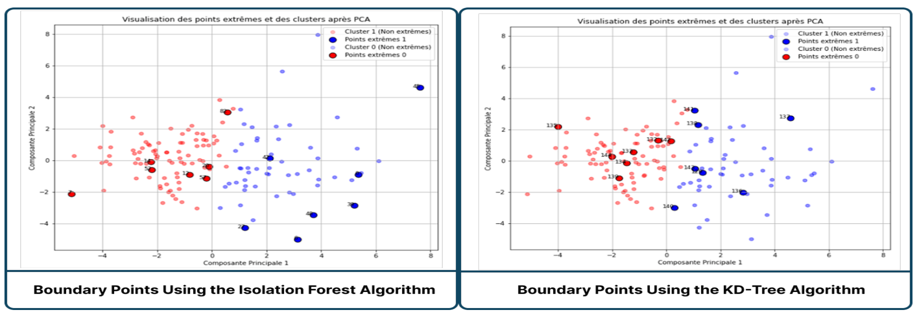

Figure 4: Cluster boundary points identified using (KD-Tree, Isolation Forest) algorithms

(c) Before identifying influential variables, we focus on boundary points. According to [10], the most relevant points to explain are those near the decision boundaries within each cluster. We therefore applied the *KD-Tree* and *Isolation Forest* methods to detect these boundary instances. A visualization of these points was performed using *PCA* (see Figure 4). Only 5% of the instances were selected as boundary points.

(d) After identifying the instances to explain, we adopted a hybrid method combining **SHAP** [11] and formal explanations generated using the (PyXAI) tool [12][7]. Our choice is detailed and justified in Section 4.4 (SHAP + Formal). This method is used to explain, for each boundary instance, why it is predicted as belonging to cluster 0 or 1 by generating rules involving the most important features.

(e) For each instance, we determine the key variables influencing the prediction. These variables are then interpreted according to their nature: either by referring to clinical thresholds for medical scores (ANX, DEP, ISI, ESS, BMI), or by comparing them to mean values for other variables.

(f) Finally, analyzing the explanations of boundary cases helps to generalize and define typical representative profiles for each cluster.

In summary, our proposed approach explains why boundary points are assigned to Cluster 0 versus Cluster 1. Analyzing these explanations enables identifying characteristic and representative profiles for each cluster.

4.4 SHAP + Formal Method

In this subsection, we introduce an approach we call **SHAP + Formal**, which combines two distinct types of XAI methods: the *SHAP* technique and formal abductive explanations (see [13, 14] for a comprehensive overview of formal methods in XAI). The *SHAP* method [11] is a feature attribution technique that quantifies the contribution of each variable to a given prediction. However, this approach has two major limitations: (1) it does not account for

[7] See `www.cril.univ-artois.fr/pyxai/` for more details about **PyXAI** and formal explanations

interactions between variables, assuming their independencean assumption that is rarely valid in practice, especially in our context, and (2) its reliability remains limited in critical domains such as medicine. Several recent studies highlight important differences between agnostic XAI methods like *SHAP* and formal approaches, see, for example, [13, 15, 16]. Furthermore, the importance scores generated by *SHAP* often fail to clearly differentiate between the identified clusters.

On the other hand, formal explanations, considered more reliable because they are based on mathematical logic [17, 13], are generally expressed in the form of rules. However, even when they are minimal, these rules can still be relatively long [18, 19] (often composed of 7 ± 2 elements), which makes them difficult to interpret [20], especially when characterizing clusters. The idea is therefore to combine the explanations provided by *SHAP*, which highlight the most influential variables, with formal explanations. Concretely, we compute the intersection between the top 10 most important variables according to *SHAP* and those present in the formal explanations, in order to obtain a rule that is simultaneously short, reliable, correct, composed of influential variables, and easy to interpret.

5 Experiments and Results

This section presents the results of the two approaches: (1) the regression-based approach and XAI to explain sleep duration, and (2) the clustering to construct patient profiles.

5.1 Regression-Based Approach and XAI to explain sleep duration

By applying the method presented in Section 3, we trained four regression models to predict sleep duration on two datasets: **Dataset_1**, covering the period between Day 0 and Day 7, and **Dataset_2**, covering the period between Day 7 and Day 17. The performances of the four models on these two datasets are presented in Table 1.

Model / Dataset	**Dataset_1**	**Dataset_2**
Linear Regression	0.66, 42.11 min	0.54, 36.38 min
Decision Tree	0.78, 48.02 min	0.70, 41.80 min
Random Forest	**0.58, 40.73 min**	**0.53, 35.85 min**
XGBoost	0.70, 43.56 min	0.62, 37.20 min

Table 1: Performance of regression models on the two datasets (MSE, MAEmin).

Based on the results in the table, the **Random Forest** model performed best on both datasets. Therefore, this model was used to predict the sleep duration of the representative instances selected according to the method described in Section 3. We then applied the SHAP explainability method to the six instances from each dataset. The SHAP explanation results were analyzed to identify the variables that contribute most to the prediction of average sleep duration, as well as whether their effects are positive (increasing sleep duration) or negative (decreasing it). Based on these analyses and the SHAP summary plot presented in Figure 5 we can get the following results :

- People who go to bed later tend to sleep more (MIDMOY).
- Regular sleep patterns promote longer sleep (SRIV).
- Insomnia (ISIV), high BMI (BMI), and excessive sleepiness (ESS) reduce sleep duration.
- Middle-aged individuals get less sleep than both younger and older adults (AGE)

5.2 Clustering to segment patients

Based on the silhouette score (see Figure 6) and the PCA visualization of the clusters for the top three values of k that yielded the best silhouette scores (as shown in Figure 7), we selected $k = 2$ as the optimal number of clusters. K-means clustering was then applied with $k = 2$, resulting in two clusters: `Cluster 0`, containing 90 instances (62.07%), and `Cluster 1`, containing 55 instances (37.93%). Subsequently, three different methods were used to characterize the clusters and identify their most discriminative variables.

*** Characterization by statistical tests.** After applying the statistical tests described in Section 4.1 to identify variables showing significant differences between the two clusters, no categorical variable was found to be statistically significant. However, *some numerical variables exhibited significant differences;* there were no significant differences between the two groups. The variables with the most significant differences are listed in Table 2.

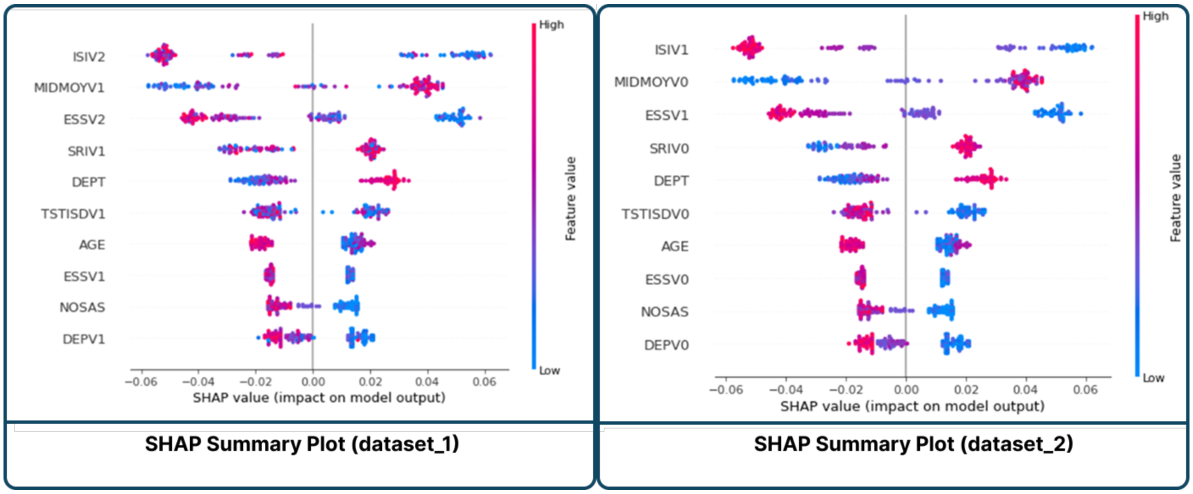

Figure 5: SHAP summary plots showing the impact of the most important variables on the prediction of sleep duration across both datasets

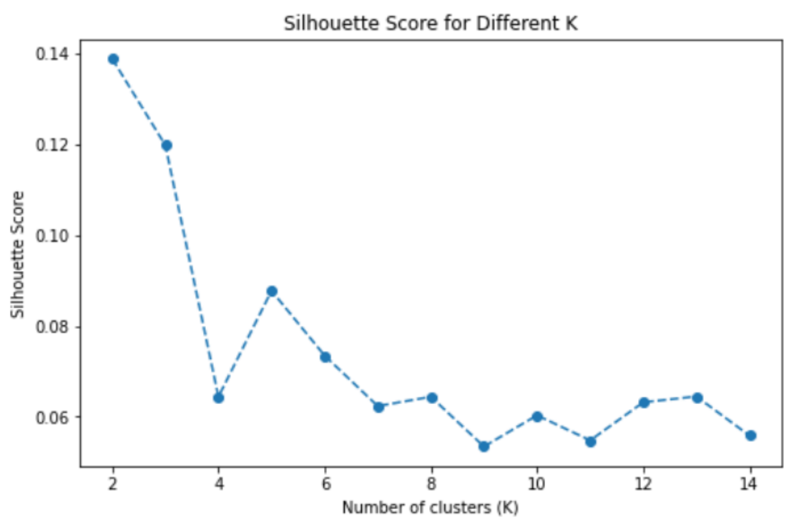

Figure 6: Silhouette Score for Different Values of k

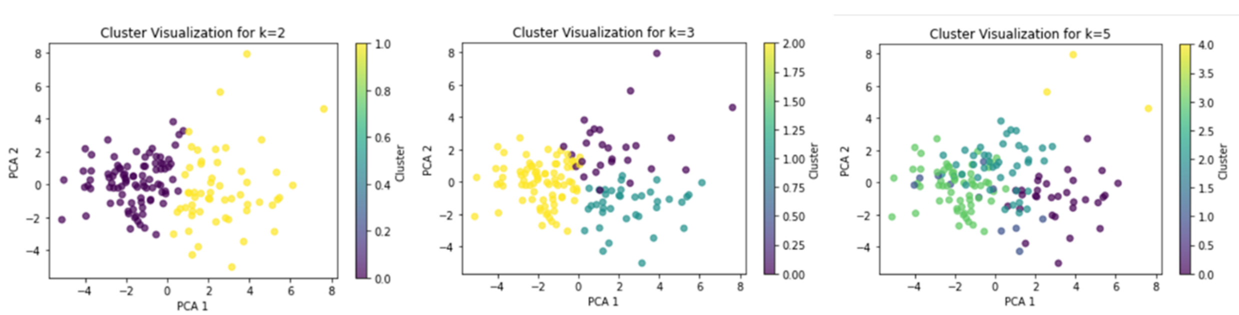

Figure 7: PCA-Based Cluster Visualization for Varying Numbers of Clusters

Analysis of Table 2 according to clinical thresholds reveals marked differences between clusters. Patients in Cluster 1 are younger (AGE) and exhibit excessive daytime sleepiness (ESS), moderate insomnia (ISI), moderate depression

Variable	Cluster 0	Cluster 1	Difference	p-value
Age	52.04 ± 12.13	47.14 ± 12.95	-4.90	2.29×10^{-2}
ESSV0	6.68 ± 4.27	10.25 ± 5.03	3.40	2.61×10^{-5}
ISIV0	13.12 ± 4.32	17.44 ± 3.68	4.31	6.84×10^{-9}
DEPV0	6.07 ± 3.37	13.25 ± 4.24	7.19	1.77×10^{-21}
ANXV0	2.52 ± 1.63	4.51 ± 1.35	1.99	3.69×10^{-12}
TSTISDV0	2.30 ± 0.92	2.87 ± 1.03	0.58	6.44×10^{-4}
SRIV0	85.59 ± 5.94	81.98 ± 7.73	-3.61	3.80×10^{-3}

Table 2: Inter-cluster comparison of significant variables (mean $\pm$ standard deviation). Positive differences indicate higher values in Cluster 1.

(DEP) with a risk of major depressive episode (ANX), as well as irregular sleep patterns (low SRIV, high variability in TSTISD and MIDISD).

In contrast, Cluster 0 groups older individuals (AGE) with normal sleepiness (ESS), mild subclinical insomnia (ISI), mild depression (DEP) without significant anxiety (ANX), and regular sleep (high SRIV and low variability in sleep parameters).

Summary. The results reveal a clear clinical dichotomy between the two clusters. Cluster 1 is characterized by a younger population exhibiting marked symptoms, combining clinical depression, moderate insomnia, and disrupted sleep regularity, which together suggest a more fragile mental and sleep health profile. In contrast, Cluster 0 includes older subjects with milder mood disorders and generally preserved sleep quality and consistency, indicating a comparatively more stable condition.

* **Distance-based characterization.** By applying the method described in Section 4.2, based on the calculation of in-patterns and out-patterns, we reproduced the approach proposed in [21], which allowed us to identify the characteristic variables of each cluster according to whether their values were generally high or low.

Cluster 0. The variables with predominantly high values are *Age*, *SRIV0*, and *SRIV1*, indicating an older population with more regular sleep. Conversely, variables with predominantly low values include *ISIV1*, *ISIV2*, *DEPV0*, *DEPV1*, *DEPV2*, *ANXV1*, and *ANXV2*, suggesting low levels of insomnia, depression, and anxiety.

Cluster 1. High values are observed for *ISIV1*, *ISIV2*, *DEPV0*, *DEPV1*, *DEPV2*, *ANXV1*, *ANXV2*, and *CRP*, indicating a high prevalence of insomnia, depression, anxiety, and possible inflammation. In contrast, *Age* is lower, indicating a younger population.

Summary. The analysis reveals two distinct profiles: Cluster 1 groups younger patients with a symptomatic triad (clinical depression, sleep disorders, and occupational activity), suggesting a potential link with work-related stress. On the other hand, Cluster 0 corresponds to older individuals with milder depression and regular sleep patterns, reflecting mental stability.

* **XAI-based characterization.** After performing clustering and adding the *Cluster* column to the dataset (see Section 4.3), we trained four classification models whose performances, optimized using **Optuna**[8], are presented in Table 3. Since the Random Forest model proved to be the most effective, it was selected for the subsequent steps.

To characterize the clusters, we determined their boundaries using the KD-Tree[9] and Isolation Forest[10] algorithms, selecting 5% of the dataset (i.e., 7 instances per cluster) as boundary points. Their distribution is visualized in Figure 4.

After identifying these boundary points, the *SHAP+Formal* explanations (see Section 4.4) revealed the key variables influencing the prediction of these boundary points. These key variables are interpreted based on standard clinical thresholds [3] when they correspond to medical scores, or by comparing them to the mean values of other variables. Below is an example of an explanation for one boundary instance from the cluster.

[8] Optuna is a hyperparameter optimization framework

[9] The kd-Tree is a structure that organizes data in a k-dimensional space (see [22]).

[10] The Isolation Forest computes an anomaly score for each instance.

Models	Accuracy	F_1 (C_0)	F_1 (C_1)
Logistic Regression	97.29%	98%	96%
Decision Tree	83.78%	88%	75%
Random Forest	**97.30%**	**98%**	**96%**
XGBoost	94.59%	96%	92%

Table 3: Comparison of classification model performances on clusters C_0 and C_1.

Cluster 0 (Instance 142). Key variables for this prediction include: DEPV0, DEPV1, DEPV2, ANXV0, ANXV1, ANXV2, ISIV0, ISIV1, TSTISDV0, and AGE. This individual shows minimal depression (DEP), no anxiety (ANX), mild subclinical insomnia (ISI), low sleep time dispersion (TSTISDV0), and is older than average.

Instance 143 (Cluster 1). Key variables for this prediction include: DEPV0, DEPV1, ISIV1, ANXV1, TSTISDV1, ESSV0, ESSV1, and AGE. This individual shows moderate depression (DEP), probable major depressive episode (ANX), marked excessive daytime sleepiness (ESS), moderate clinical insomnia (ISI), high sleep time dispersion (TSTISDV1), and is younger than average.

Summary. The interpretation of boundary instances between the two clusters reveals that individuals in Cluster 0 are generally older, less depressed, and exhibit regular sleep patterns. In contrast, those in Cluster 1 tend to be younger, suffer from depression and insomnia, and have irregular and insufficient sleep.

6 Related work

In recent years, numerous studies have applied machine learning (ML) techniques to better understand sleep behaviors and detect sleep disorders. These works rely on supervised and unsupervised learning methods, often trained using data from wearable devices, mobile applications, or clinical cohorts.

Trujillo et al. [23] proposed a regression model to predict sleep duration from physical activity and sleep data collected over a 30-day period, identifying exercise and sleep regularity as key predictors. Al-Mamun et al. [24] used a *CatBoost* classifier to identify abnormal sleep duration in students, based on self-reported demographic and psychological variables. Although both approaches focus on supervised learning, they do not explicitly address model interpretability.

Unsupervised approaches, such as those by Park et al. [25] and Ferreira-Santos and Rodrigues [26], have used clustering to identify sleep-related subtypes, often applying deep learning or categorical clustering techniques to identify profiles in patients with insomnia or sleep apnea. Similarly, Gool et al. [27] used hierarchical clustering to define subtypes in hypersomnolence disorders. These approaches provide population-level analyses but generally lack individualized and interpretable characterizations.

Positioning of our approach *w.r.t.* related work. Our contribution builds on these works by combining supervised regression and unsupervised clustering while introducing explainability techniques at each stage. Unlike previous studies, we not only identify predictors of sleep duration but also characterize latent groups using explainable AI (XAI) tools. Our method enables both a global analysis and a rigorous interpretation at the individual level via the SHAP+Formal method and rule-based profiles. This hybrid and interpretable design enhances the usability of ML + XAI in sleep disorder research, particularly for clinicians and non-expert users.

7 Conclusion

Our regression analysis identified several variables associated with sleep duration. Individuals who go to bed later tend to sleep longer, while regular sleep patterns are linked to a longer duration. Conversely, a high BMI, insomnia, and daytime sleepiness are associated with reduced sleep time. Middle-aged adults also sleep less than younger and older individuals.

The clustering approach revealed two distinct patient profiles. The three characterization methods produced similar results, highlighting:

- A group of young individuals suffering from depression and insomnia, with irregular and insufficient sleep.

- A group of older, less depressed individuals with regular and sufficient sleep.

While conventional methods identify key variables per group, our XAI-based approach provides deeper analyses. It allows visualizing the contribution of each variable and extracting rule-based explanations, particularly for borderline cases. This strengthens interpretability and trust in the clustering process.

Future directions. One future research direction involves improving missing value handling. Our explainable framework could be extended to incorporate imputation techniques while preserving interpretability. This would allow models to remain transparent and reliable in more realistic and incomplete data contexts.

Reproducibility. To encourage reproducibility and facilitate further research, the source code used in this study will be made publicly available on GitHub under an open-source license once the article is published.

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